

Problem 5

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**Problem 1** The volume of a doughnut with an inner radius of  $a$  and an outer radius of  $b$  is

$$V = \pi^2 \frac{(b+a)(b-a)^2}{4}.$$

Find  $db/da$  if the volume of a doughnut is  $64\pi^2$  and does not change.